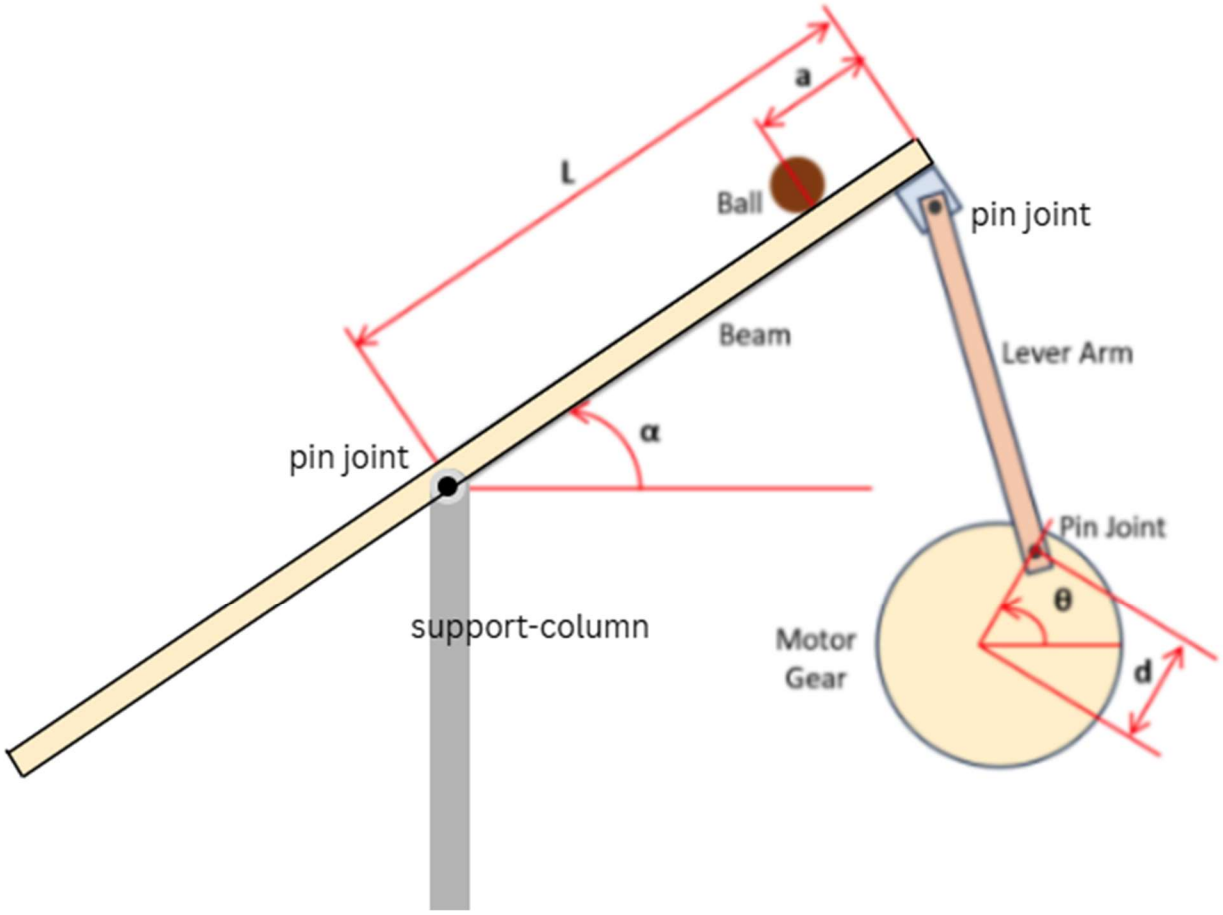
Course Project: Part-II

ES 245: Fall 2025 Term

Problem Statement

As shown in the figure below, a ball (or Car) is placed on a beam, where it rolls/slides under the influence of gravity. The ball/car has one degree of freedom along the length of the beam. The beam's angle is adjusted using a four-bar linkage, with servo motor gear connected to a lever arm or a coupler link connecting the beam. As the servo motor gear rotates by an angle θ about its axis, the lever arm angle changes by α from the horizontal.



- **Input (Control Signal):** Servo angle 'θ'.
- **Output (Measured Signal):** Ball position "a", sensed by a proximity (**Ultrasonic, Analog IR, etc.)** sensor placed along the beam.
- **General Hardware:** The setup includes a servo motor with four bar type linkage to tilt the beam, a beam with free-rolling ball, or an alternative rolling/sliding object, and a proximity sensor to detect the ball's position. Arduino Uno/Mega will be provided for testing the controller.

Part 2: Frequency Domain Analysis

Task 5: System Characterization [10 marks]

- Perform system characterization by applying a frequency-varying actuator (chirp) signal and recording the corresponding sensor response. (The code for this will be provided).
- Derive the transfer function of the system from the input-output relationship. (use n4sid in MATLAB)
- Plot the Bode plot and analyze the system's gain and phase behavior.

Task 6: Root Locus and Frequency Response Analysis of the Ball/Car–Beam System [15 marks]

6.1 Root Locus and Nyquist Analysis [5]

- Using MATLAB, plot the following for the system's open-loop transfer function:
    1. Root Locus – Comment on the stability and dominant poles of the system.
    2. Nyquist Plot – Analyze stability margins and system robustness using the Nyquist stability criterion.

6.2 Bode Plot Analysis [5]

- Plot the Bode diagram for the open-loop transfer function.
- Report and interpret the gain margin and phase margin values.
- Compare and relate the frequency-domain stability margins with observations from the Root Locus and Nyquist plots.

6.3 Design Criteria [5]

- Evaluate system performance with respect to:
    - Overshoot < 5%
    - Settling time < 5 seconds (within a 5% tolerance band).
- Use these criteria to assess the adequacy of the existing system response and identify the need for compensator improvement (to be designed in later tasks).

Task 7: Compensator Identification – Lead or Lag [10 marks]

- Based on the frequency-domain characteristics obtained from Task 6, identify and design the required Lead or Lag compensator to achieve the desired transient and steady-state performance.
- Verify improvements in system response using updated Bode plots and time-domain step responses.

Task 8: Hardware Implementation [15 marks]

- Implement and tune the designed controllers on the provided hardware setups
- Demonstrate stable control performance and response to small perturbations.
- Submit a detailed project report including modeling, system characterization, analysis plots, compensator design, simulation results, and hardware implementation findings.